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How Well Do Phonological Awareness and Rapid Automatized Naming Correlate With Chinese Reading Accuracy and Fluency? A Meta-Analysis

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ABSTRACT

Previous meta-analyses on the relationship between phonological awareness, rapid automatized naming (RAN), and reading have been conducted primarily in English, an atypical alphabetic orthography. Here, we aimed to examine the association between phonological awareness, RAN, and word reading in a nonalphabetic language (Chinese). A random-effects model analysis of data from 35 studies revealed a moderate relationship of phonological awareness with reading accuracy ($r = .36$) and reading fluency ($r = .39$). RAN also correlated significantly with reading accuracy ($r = -.38$) and reading fluency ($r = -.51$), but its relationship varied as a function of test type (graphological RAN correlated more strongly with reading than nongraphological RAN) and reading outcome (RAN correlated more strongly with reading fluency than reading accuracy). Age/grade and dialect (Mandarin vs. Cantonese) did not influence the size of the correlations. Taken together, the findings of this meta-analysis suggest that phonological awareness and RAN are universal correlates of word reading.

Several studies have established that phonological awareness and rapid automatized naming (RAN) are significant predictors of reading across languages (e.g., Compton, 2003; Georgiou, Parrila, Cui, & Papadopoulos, 2013; Moll, Fussenegger, Willburger, & Landerl, 2009; Parrila, Kirby, & McQuarrie, 2004; Warmington & Hulme, 2012; Yeung et al., 2011; Ziegler et al., 2010) and core deficits in dyslexia (e.g., Furnes & Samuelsson, 2010; Ho, Chan, Lee, Tsang, & Luan, 2004; Kirby, Parrila, & Pfeiffer, 2003; Landerl et al., 2013; Wolf & Bowers, 1999). Because of their relatively independent role in reading, Wolf and Bowers (1999) proposed the double-deficit hypothesis, according to which children with deficits in both phonological awareness and RAN will experience more difficulties in reading than children with single deficits in either phonological awareness or RAN—a hypothesis that has been confirmed in different languages (e.g., English: Kirby et al., 2003; Greek: Papadopoulos, Georgiou, & Kendeou, 2009; Finnish: Torppa, Georgiou, Salmi, Eklund, & Lyytinen, 2012; Chinese: McBride-Chang & Ho, 2000b).

The size of the relationship between phonological awareness, RAN, and word reading has also been estimated in a few meta-analyses. Scarborough (1998) and Swanson, Trainin, Necochea, and Hammill (2003) found that the average correlation between phonological awareness and word reading was .46 and .41, respectively. In turn, the average correlation between RAN and word reading was .38 and .41, respectively. Unfortunately, both meta-analyses included studies conducted predominantly in English, an atypical alphabetic orthography (Share, 2008). Two more meta-analyses have examined the effects of either phonological awareness (Melby-Lervåg, Lyster, & Hulme, 2012) or RAN (Araújo, Reis, Petersson, & Faisca, 2015) on reading. Melby-Lervåg et al.

(2012) reported an average correlation of .57 between phonological awareness and word reading and Araújo et al. (2015) an average correlation of .45 between RAN and word reading. Melby-Lervåg et al.'s meta-analysis included studies conducted only in alphabetic languages, and as a result we do not know if their findings generalize to nonalphabetic languages, such as Chinese. In turn, Araújo et al. contrasted the role of RAN in alphabetic and nonalphabetic languages, but did so by including studies conducted in Japanese, Chinese, and Korean in the same group. Given that these orthographies vary from each other on several important linguistic aspects (Taylor & Taylor, 2014; Wang & Tsai, 2011) and RAN has been found to predict reading differently in these orthographies (Cho & Chiu, 2015), merging the results across these studies likely inflated RAN's contribution to reading in nonalphabetic orthographies. Thus, in this meta-analysis we aimed to estimate the size of the relationship between phonological awareness, RAN, and word-reading accuracy and fluency in a nonalphabetic language (Chinese).

A plethora of studies has shown that there is a close and probably causal relationship between phonological awareness and word reading in English and other alphabetic languages (e.g., Caravolas et al., 2012; Moll et al., 2014; Ziegler et al., 2010). Obviously, the ability to identify and manipulate the individual sounds in one's oral language enables children to map these sounds (phonemes) to graphemes (letters), which is quintessential for learning to read (Ehri, 2005). However, the relationship between phonology and orthography is radically different in Chinese than in alphabetic languages (see Hanley, 2005). Given that phonemes are not directly represented in the Chinese writing system, one might expect that the relationship between phonological awareness and learning to read will be weaker in Chinese than in alphabetic languages.

In support of this hypothesis, some researchers have argued that phonological awareness plays a rather limited role in learning to read Chinese compared to that played in alphabetic languages (e.g., Chow, McBride-Chang, & Burgess, 2005; Huang & Hanley, 1995, 1997; McBride-Chang et al., 2005; McBride-Chang & Ho, 2000a; Shu, McBride-Chang, Wu, & Liu, 2006; Taylor, 2002). Chinese is a morphosyllabic language in which the basic unit, the character, represents a syllable and a morpheme. Thus, children do not need to decompose syllables into phonemes in order to accurately read Chinese. In addition, although the phonetic radical that is used in about 80% of modern Chinese characters provides a clue to the character's pronunciation, it is relatively ambiguous and, in some cases, even misleading (e.g., Shu, Chen, Anderson, Wu, & Xuan, 2003). It has been estimated that only 23% to 26% of the phonetic compound characters (when tone is taken into account) can be read accurately using the phonetic radical (Chung & Leung, 2008; however, Zhou, 1978, estimated it to be 39%).

However, because the phonetic information in Chinese characters is encoded at the syllabic level, some researchers have argued that the ability to dissect words into syllables and syllables into onsets and rimes might be important in learning to read Chinese (e.g., Ho & Bryant, 1997; Hu & Catts, 1998; McBride-Chang, Bialystok, Chong, & Li, 2004; Shu, Peng, & McBride-Chang, 2008; Siok & Fletcher, 2001). Several studies have shown that syllabic awareness (including onset and rime awareness) predicts character recognition, particularly in preschool and early grades (e.g., Ho & Bryant, 1997; McBride-Chang et al., 2004; Pan et al., 2011; Tong et al., 2011; Wei et al., 2014; Zhang et al., 2013). Taken together, this suggests that the type of phonological awareness tasks (simple: e.g., syllable deletion vs. complex: e.g., phoneme deletion) may influence the size of the correlations with word reading.

Some studies have also shown that Chinese readers who have been exposed to a phonetic coding system (e.g., Pinyin in mainland China and Zhuyin Fuhao in Taiwan) develop better phonological awareness skills than Chinese readers who have not (Huang & Hanley, 1995; Leong, Cheng, & Tan, 2005; McBride-Chang et al., 2004). Siok and Fletcher (2001) further argued that the effects of phonological awareness in Chinese reading may be mediated by Pinyin skills. Thus, exposure or not of Chinese children to a phonetic coding system may be another important dimension of the phonological awareness–reading relationship in Chinese that needs to be examined more closely.

Several studies have also shown that RAN is a significant predictor of Chinese reading (e.g., Chow et al., 2005; Hu & Catts, 1998; Liao, Georgiou, & Parrila, 2008; Luo, Chen, Deacon, Zhang, & Yin, 2013; McBride-Chang & Ho, 2005; McBride-Chang & Kail, 2002; Pan et al., 2011; Wei et al., 2014; Wei, Georgiou, & Deng, in press; Yeung et al., 2011) and that its contribution is independent of phonological awareness (e.g., Ho et al., 2004; Liao et al., 2015; McBride-Chang & Ho, 2000b; Pan et al., 2011; Xue, Shu, Li, & Tian, 2013; Yeung et al., 2011).

Researchers have attributed RAN's connection to Chinese reading to at least two reasons: First, because Chinese involves many visually complex characters, visual processing skills have been found to predict Chinese reading (e.g., Huang & Hanley, 1995; Luo et al., 2013; Siok & Fletcher, 2001). Because RAN also requires visual processing of stimuli (Stainthorp, Stuart, Powell, Quinlan, & Garwood, 2010; Wolf & Bowers, 1999; see also Arnell, Joanisse, Klein, Busseri, & Tannock, 2009, for findings showing that visual processing skills predict common variance between RAN and reading), it should also predict Chinese reading. Because the Chinese characters are far more complex than alphabetic letters, if this hypothesis is true, it might be reasonable to predict a much stronger relationship between RAN and Chinese reading than between RAN and reading in alphabetic languages (and perhaps equally strong relationships between RAN and reading in English and in transparent orthographies, as letters in different alphabetic scripts are of similar visual complexity). Araújo et al.'s (2015) study partly supported this hypothesis because the correlation between RAN and reading fluency (but not reading accuracy) in nonalphabetic orthographies ($r = .54$) was significantly stronger than the corresponding one in alphabetic orthographies ($r = .48$).

Second, because of the relatively large number of ambiguous symbol-sound correspondences in Chinese, Chinese characters are generally taught by rote memorization (Hanley, 2005). As a result, children are required to become efficient in making relatively arbitrary connections between symbols and sounds. According to Manis, Seidenberg, and Doi (1999), arbitrariness is also central in RAN because the visual stimuli in RAN tasks have to be mapped rapidly to their names, and these mappings are arbitrary. For example, seeing the digit “5” does not equip the participant with the phonological information needed to say the word “five.” In the same way, producing the correct pronunciation for an exception word (i.e., *yacht*) requires the retrieval of partially arbitrary item-specific knowledge. Thus, RAN should be important in learning to read Chinese. However, if the “arbitrary-connection” hypothesis is true, we should observe higher correlations between RAN and reading in Chinese than in English and higher correlations in English than in more transparent orthographies (e.g., Finnish, Greek). Georgiou and colleagues (Georgiou, Aro, Liao, & Parrila, 2015; Georgiou, Parrila, & Liao, 2008) directly tested this hypothesis by comparing the size of the correlations in Chinese and alphabetic orthographies. The results of both studies showed no significant differences in the RAN–reading relationship across languages.

However, several questions remain unanswered regarding RAN's relationship with Chinese reading. First, we do not know if the relationship between RAN and Chinese reading varies as a function of reading outcome (reading accuracy vs. reading fluency). Scarborough (1998) and Swanson et al. (2003) did not address this question, and Araújo and colleagues (2015) combined Chinese with Korean and Japanese studies, which does not allow us to draw any firm conclusions on the RAN–reading fluency relationship in Chinese. Although some studies in Chinese have reported stronger correlations between RAN and reading fluency than reading accuracy (e.g., Georgiou et al., 2015; Wei et al., in press; Xue et al., 2013), some others have reported lower correlations (e.g., Ding, Richman, Yang, & Guo, 2010; Kidd, Shum, Ho, & Au, 2015). From a theoretical point of view, if RAN is a “microcosm” of the reading fluency system (Norton & Wolf, 2012; Protopapas, Altani, & Georgiou, 2013), it should also produce significantly stronger correlations with reading fluency than with reading accuracy. Second, it remains unclear if the relationship between RAN and Chinese reading varies as a function of RAN task. Scarborough and Araújo et al. reported higher correlations between graphological RAN (digits and letters) and reading than between nongraphological RAN

(colors and objects) and reading. However, some studies in Chinese have shown that Object Naming correlates equally well with Character Recognition as Digit or Character Naming (e.g., Ding et al., 2010; McBride-Chang, Shu, Zhou, Wat, & Wagner, 2003). Because Chinese characters convey meaning and Object Naming involves access to semantic information (e.g., Cummine, Szepesvari, Chouinard, Hanif, & Georgiou, 2014), it is possible that, in Chinese, nongraphological RAN correlates equally well with word reading as graphological RAN. Third, we do not know if the relationship between RAN and reading varies as a function of Chinese dialect (Mandarin vs. Cantonese). Mandarin- and Cantonese-speaking children learn two different Chinese scripts, simplified and traditional. Compared to the simplified script that is used in mainland China, traditional characters (used in Hong Kong) have more strokes and visual features. In addition, whereas in mainland China and Taiwan children learn new characters with the aid of a phonetic coding system (e.g., Pinyin), Cantonese-speaking children learn to recognize new characters using the principle of “look and say.” This has an important implication regarding the role of RAN in Chinese reading. If RAN is related to Chinese reading because they both involve making arbitrary connections between symbols and sounds (Manis et al., 1999), then the associations should be higher in Cantonese than in Mandarin, because in Mandarin the use of Pinyin or Zhuyin Fuhao makes learning of new characters less arbitrary. Finally, it is not clear whether age or grade level influence the RAN–reading relationship. In their meta-analysis, Araújo et al. (2015) found that grade level influenced the correlations only when reading accuracy was the outcome measure (RAN was more strongly related to word reading in beginning readers) and only in opaque orthographies. Certainly, this finding needs to be replicated.

The present study

The purpose of this meta-analysis was to examine the strength of the relationship between phonological awareness, RAN, and word-reading accuracy and fluency in Chinese. More specifically, we aimed to answer the following questions:

- (1) What is the magnitude of the association between phonological awareness and Chinese reading?
- (2) What is the magnitude of the association between RAN and Chinese reading?
- (3) Do the relationships between phonological awareness, RAN, and Chinese reading vary as a function (a) of the tasks used to assess phonological awareness (simple vs. complex), RAN (graphological vs. nongraphological), and reading (accuracy vs. fluency); (b) of the dialect (Mandarin vs. Cantonese); and (c) of children’s age or grade?

Method

Data collection

The inclusion, search, and coding procedures are detailed in Figure 1. For the target constructs examined in our study (phonological awareness, RAN, reading accuracy, and reading fluency), operational criteria were established to determine the indicators of each construct. A task was considered to be a measure of phonological awareness if it involved manipulation, generation, or judgment of phoneme, onset, rhymes, syllables, or lexical tone in words. In turn, to be considered a measure of RAN, the test should require quick naming of an array of objects, colors, letters, digits, or other symbols in Chinese, such as Chinese characters, Pinyin, or Zhuyin Fuhao. Reading accuracy included measures requiring accurate Chinese word recognition without imposing any time limits. Finally, to be considered a measure of reading fluency, the task should require children to read as many Chinese characters, words, or sentences as possible within a specified time limit.

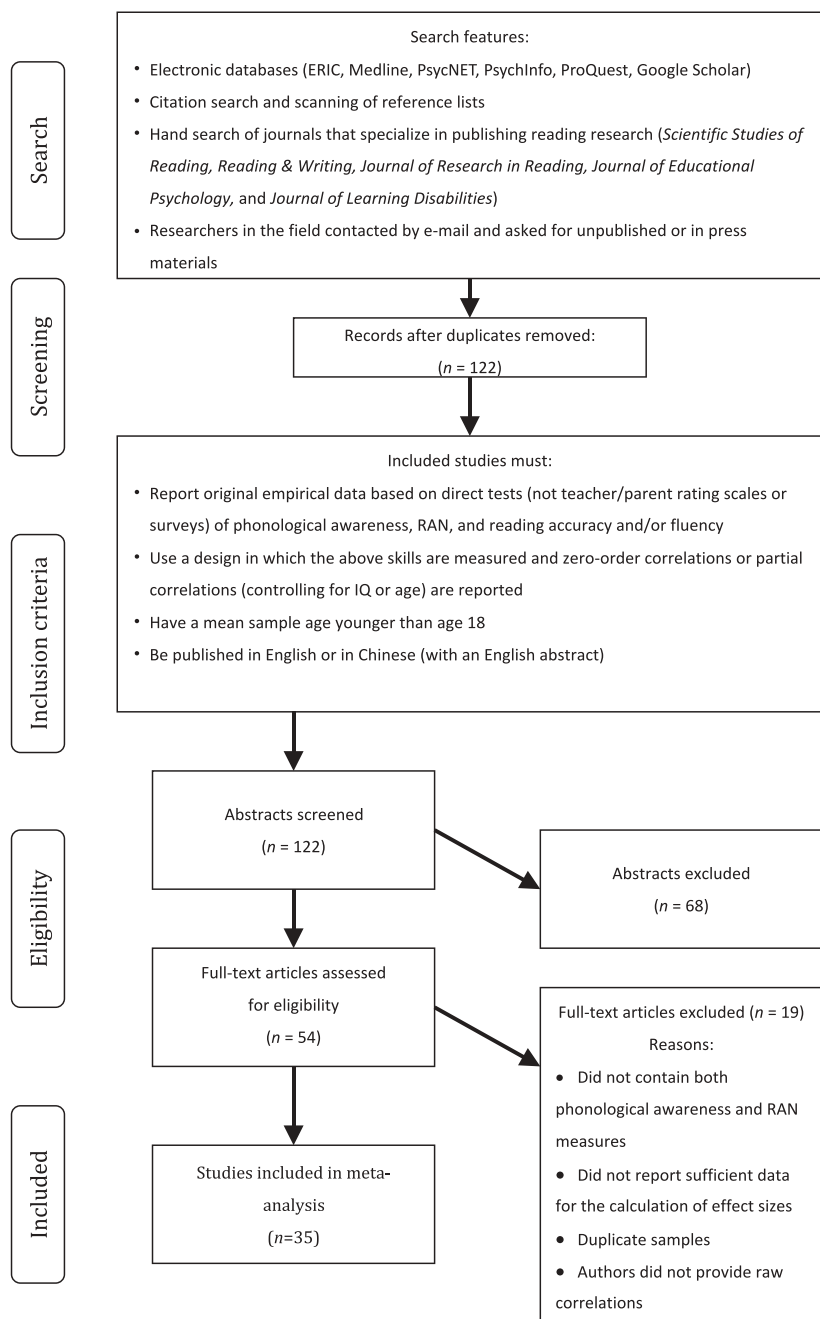


Figure 1. Flow diagram for the search and inclusion of studies (adapted from <http://www.prisma-statement.org>).

A search of abstracts was conducted in the computerized databases (ERIC, Medline, PsycNET, PsycINFO, ProQuest, and Google Scholar) for publications in both English and Chinese from January 1990 to December 2014 and with the descriptors Chinese, China, Hong Kong, Taiwan paired with phon* awareness, phonological processing, RAN, rapid automatized naming, naming speed, reading, and character recognition. Abstracts of peer-reviewed studies, dissertations, and book chapters were subsequently scrutinized. Similar to Swanson et al. (2003), only studies that assessed

both phonological awareness and RAN were included in our meta-analysis. The final sample included 35 studies, with 52 different samples, comprising a total of 6,809 Chinese children.¹

To prevent violation of independence of observations (i.e., including data from studies that used the same sample more than once), studies by the same author were examined in order to detect duplicate samples. For longitudinal studies, the first time point in which reading ability was measured was coded. The possibility of coding longitudinal correlations was also considered, but there were only a few comparable longitudinal studies ($n = 9$) for a meta-analysis on longitudinal correlations to be performed.

Because some studies used multiple measures for each of the target construct, a set of rules was established to help us determine which indicator should be coded. For phonological awareness, phoneme-based and lexical tone measures were coded before other types of phonological awareness measures, such as awareness of larger units (i.e., rhymes or syllables) or composite scores, because tasks measuring awareness of larger units have been found to be sensitive only in younger children (e.g., Li, Shu, McBride-Chang, Liu, & Peng, 2012; Siok & Fletcher, 2001). For RAN, graphological (digits and characters) tasks were coded before nongraphological (colors and objects) tasks because graphological RAN tasks require print to sound conversions, and in this sense they are closer to reading than nongraphological RAN tasks that also involve the extra step of semantic processing (e.g., Cummine et al., 2014). If more than one graphological RAN task was used in a study, the average correlation was coded. For reading accuracy and reading fluency, single character reading or compound word reading was coded before text reading because they are considered “purer” measures of word reading than text reading that also engages syntactic, grammatical, and semantic processing.

All studies were coded twice by trained coders, and interrater reliability was estimated. The interrater correlation (Pearson’s) for the reported correlation coefficients was .99, and the agreement rate was 97%. In regards to the moderator variables (see next), the interrater correlation for age was .99 and the agreement rate 96%. Cohen’s K for the categorical moderator variables (dialect and test type) was .97. The few disagreements in the coding were resolved after a discussion between the coders and the second author.

Moderator variables

We coded three important moderators that could explain variation between the studies. When fewer than five studies reported data on a moderator, the analysis for this moderator was not performed.

Age/grade

Age was coded in the following way: When age range and mean age were both reported and age range was not larger than 1 year, the mean age was coded (e.g., when age range was 4.5–5.4 years and mean age was 5.02 years, age was code as 5.02 years). When only age range was reported and the range was not larger than one year, the median value of the range was coded (e.g., when age range was 4.5–5.4 years and mean age was not reported, 4.95 years was coded as age). When only the mean age was reported, then the mean age was coded. Finally, studies in which only the age range was

¹Because some of our studies included also correlations between morphological awareness, orthographic knowledge, and Chinese reading, we also calculated the average correlation for these linguistic skills. Twenty-one studies reported correlations between morphological awareness and reading accuracy ($N = 3,541$, M sample size = 168.62, $SD = 132.01$, range = 47–496, age range = 5:1–12:2 years). The overall mean correlation was significant, $r = .401$, 95% confidence interval (CI) [.346, .453], $z(20) = 22.907$, $p < .001$. Only two of these studies reported also correlations with reading fluency with the overall mean correlation being marginally nonsignificant, $r = .283$, 95% CI [–.004, .527], $z(1) = 3.684$, $p = .053$.

Eighteen studies reported correlations between orthographic knowledge and reading accuracy, and three of them also reported correlations with reading fluency. The overall mean correlation for reading accuracy was significant, $r = .305$, 95% CI [.225, .380], $z(17) = 7.206$, $p < .001$ ($N = 3,227$, M sample size = 179.28, $SD = 140.36$, range = 53–496, age range = 5:4–12:2 years). The overall mean correlation for reading fluency was also significant, $r = .262$, 95% CI [.128, .387], $z(2) = 3.778$, $p < .001$ ($N = 207$, M sample size = 69, $SD = 18.73$, range = 54–90, age range = 8:0–10:6 years).

reported and the range was larger than one year were excluded from the moderator analyses (e.g., when age range was 3.5–6.7 years and mean age was not reported). In addition, we coded grade and used it as a moderator variable. Only studies in which the participants were recruited from a single grade were included in the moderator analyses.

Test type

Phonological awareness. Phonological awareness tasks that required judgment or manipulation of lexical tone (e.g., which of the following Chinese characters is pronounced in a different tone than the other two? /re4/, /fan1/, or /ben1/) or phoneme (e.g., what is left if we delete the /m/ sound from /mei4/?) were classified as complex tasks, whereas others tasks, such as rime detection (e.g., which Chinese word does not rime with the other two? /zhu1/, /wa1/, or /ya1/) and syllable deletion (e.g., what is left if we delete /shu1/ from /shu1bao1/?) were classified as simple ones.

RAN. Tasks were classified into two categories: graphological (digits and characters) and nongraphological (colors and objects).

Dialect. We used two categories for dialect: Mandarin (spoken in mainland China and Taiwan) and Cantonese (spoken in Hong Kong and Guangzhou).

Meta-analytic procedures

The computer program “comprehensive meta-analysis” (Borenstein, Hedges, Higgins, & Rothstein, 2005) was used for the majority of the analyses. Correlations between the main target constructs (phonological awareness and RAN) and word reading (accuracy and fluency), as well as information pertinent to age/grade, test type, and dialect were coded from the studies and entered into a predefined data sheet in the computer program (see Table A1 in the appendix for information on the studies included in our meta-analysis).

The effect sizes for the studies were displayed by the correlation coefficient Pearson’s r . A 95% CI was calculated for each effect size to examine whether the correlation was differently different from zero. The overall correlation was estimated by calculating a weighted average of the correlations from each study. We used a random-effects model, which rests on the assumption that variation between studies can be systematic and not only due to random error. Whether the overall correlation differed from zero was tested with a z test. A sensitivity analysis was also conducted to examine the impact on the overall range of correlations when outliers were removed. Values were removed one at a time in order to calculate a new overall correlation, and the range of this new overall correlation was checked again to make sure the overall correlation was stable.

To examine whether the variation in correlations between studies was significant, the Q test of homogeneity was used (Hedges & Olkin, 2014). A significant value on this test indicates a reliable variability between the correlations in the sample of studies. I^2 was used to determine the magnitude of heterogeneity. I^2 is the proportion of total variation between the correlations that is caused by real heterogeneity rather than chance.

A metaregression based on method of moments for random-effects models was used to predict study outcome from the moderator variables. To determine the strength of the predictors, the proportion of between-study variance explained (R^2) was used as an effect size. All computations of the metaregression were carried out by macros developed in SPSS (Wilson, 2005). For the categorical moderator variables (dialect and test type), the studies were separated in subsets based on the categories of the moderator variable. The degree of differences between the subsets of studies was tested with a Q test and by comparing the correlation magnitude with CIs between the study subsets.

A funnel plot for random-effects models was used to determine the presence of retrieval bias. In the funnel plot, sample size is plotted on the y axis and effect size on the x axis. In the absence of

retrieval bias, this plot should be expected to form an inverted funnel. In the presence of bias, the funnel will be asymmetric. To detect retrieval bias, funnel plots are examined for all analyses presented. The “trim and fill” for random-effects models (Duval & Tweedie, 2000) was used to examine the impact from possible missing studies. The “trim and fill” method imputes values in the funnel plot to make it symmetrical and calculate an estimated overall effect size on this basis.

Results

Correlations between phonological awareness and reading accuracy

Fifty-one independent correlations comprising 6,598 children (M sample size = 129.37, SD = 115.69, range = 26–518) examined the relationship between phonological awareness and reading accuracy. The age of the samples ranged from 3:4 to 12:4 years. As shown in Figure 2, the overall mean correlation was moderate, $r = .365$, 95% CI [.323, .404], and significant, $z(50) = 16.388$, $p < .001$. The variation in correlations between studies was significant and large, $Q(51) = 160.392$, $p < .001$, $I^2 = 68.826$. After removing outliers in the sensitivity analysis, the overall correlation ranged from $r = .361$, 95% CI [.323, .398], to $r = .376$, 95% CI [.337, .414]. The funnel plot indicated no publication bias in the results. As shown in Table 1, neither test type nor dialect reliably explained variation in the correlation magnitude. Finally, a separate analysis examining the effects of age

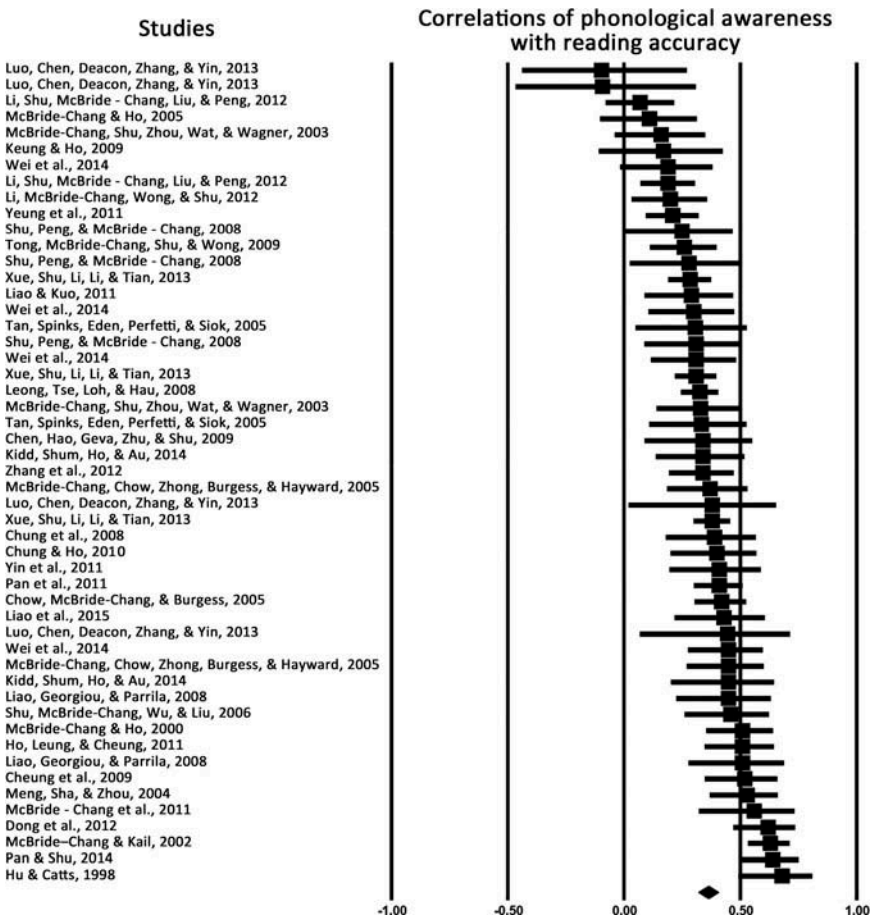


Figure 2. Overall average correlation (displayed by ♦) and correlation with confidence interval for each study correlating phonological awareness and reading accuracy.

Table 1. Number of correlations, correlation size, 95% CI, heterogeneity statistics, difference in *r* between categories and significance test of categories for moderators of the relationship between phonological awareness and reading accuracy.

Moderator variable	No. of correlations (k)	Correlation (r)	95% CI	Heterogeneity (<i>I</i> ²)	Difference in <i>r</i> (highest–lowest category)	Significance test of difference (Q test)
Test type						
Simple	38	.346**	.29, .40	73.63**		
Complex	27	.347**	.30, .40	59.77**	.001	.967
Dialect						
Cantonese	19	.359**	.29, .42	72.26**		
Mandarin	32	.377**	.33, .43	66.17**	.018	.654

Note. CI = confidence interval; *I*² = the proportion of total variation between the effect sizes that are caused by real heterogeneity rather than chance; *r* = correlation size (Pearson's *r*) for subsets of studies belonging to different categories of the moderator variable
***p* < .01.

indicated that it did not reliably explain variation in the size of the correlations, $\beta = -0.008$, $p = .969$, $k = 27$, $R^2 < .001$. This means that the correlations between phonological awareness and reading accuracy were stable across the age range (here 3:7–12:2 years). Using grade instead of age did not reliably explain variation in the size of the correlations either, $\beta = 0.128$, $p = .542$, $k = 24$, $R^2 = 0.016$.

Correlations between phonological awareness and reading fluency

Thirteen independent correlations comprising 1,174 children (M sample size = 90.31, $SD = 54.43$, range = 47–262) examined the relationship between phonological awareness and reading fluency. The age of the samples ranged from 5:1 to 12:4 years. As shown in Figure 3, the overall mean correlation was moderate, $r = .391$, 95% CI [.307, .468], and significant, $z(12) = 8.53$, $p < .001$. The variation in correlations between studies was significant and large, $Q(12) = 29.644$, $p < .001$, $I^2 = 59.519$. After removing outliers in the sensitivity analysis, the overall correlation ranged from $r = .362$, 95% CI [.296, .424], to $r = .410$, 95% CI [.332, .483]. The funnel plot indicated no publication bias in the results. As shown in Table 2, neither test type nor dialect reliably explained variation in the correlation magnitude. Furthermore, neither age ($\beta = 0.122$, $p = .878$, $k = 4$, $R^2 = 0.015$) nor grade ($\beta = -0.481$, $p = .275$, $k = 6$, $R^2 = 0.231$) reliably explained variation in the size of the correlations. This means that the correlations between phonological and reading fluency were stable across the age (here, 7:5–8:9 years)

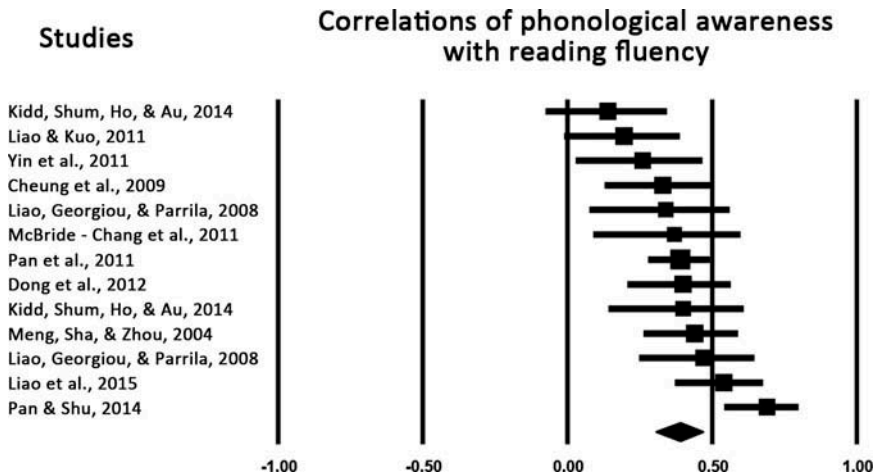


Figure 3. Overall average correlation (displayed by ♦) and correlation with confidence interval for each study correlating phonological awareness and reading fluency.

Table 2. Number of correlations, correlation size, 95% CI, heterogeneity statistics, difference in r between categories and significance test of categories for moderators of the relationship between phonological awareness and reading fluency.

Moderator variable	No. of correlations (k)	Correlation (r)	95% CI	Heterogeneity (I^2)	Difference in r (highest–lowest category)	Significance test of difference (Q test)
Test type						
Simple	10	.335**	.21, .45	73.73**		
Complex	6	.409**	.32, .49	5.03	.074	.324
Dialect						
Cantonese	3	.282**	.13, .43	33.24		
Mandarin	10	.419**	.33, .50	59.66**	.137	.114

Note. CI = confidence interval; I^2 = the proportion of total variation between the effect sizes that are caused by real heterogeneity rather than chance; r = correlation size (Pearson's r) for subsets of studies belonging to different categories of the moderator variable.

** $p < .01$.

or grade (here, Grades 1–6) range. Finally, phonological awareness did not correlate more strongly with reading fluency than reading accuracy, $Q(1) = 0.315$, $p = .575$.

Correlations between RAN and reading accuracy

Fifty independent correlations comprising 6,783 children (M sample size = 135.66, $SD = 114.23$, range = 47–518) examined the relationship between RAN and reading accuracy. The age of the samples ranged from 3:4 to 12:4 years. As shown in Figure 4, the overall mean correlation was moderate, $r = -.387$, 95% CI $[-.418, -.355]$, and significant, $z(49) = -21.494$, $p < .001$. The variation in correlations between studies was significant and large, $Q(51) = 102.634$, $p < .001$, $I^2 = 52.258$. After removing outliers in the sensitivity analysis, the overall correlation ranged from $r = -.391$, 95% CI $[-.422, -.360]$, to $r = -.381$, 95% CI $[-.411, -.350]$. In addition, the funnel plot indicated that studies were missing to the right of the mean (i.e., studies with correlations above the overall mean). In a “trim and fill” analysis two studies were added and the adjusted overall correlation was $r = -.363$, 95% CI $[-.40, -.33]$. As shown in Table 3, test type was a significant moderator variable, whereas dialect did not reliably explain variation in the correlation magnitude. Studies using graphological RAN tests produced significantly larger correlations with reading accuracy than studies using nongraphological RAN tests. Finally, a separate analysis examining the effects of age or grade indicated that they did not have a significant impact on the size of the correlations (Age: $\beta = 0.176$, $p = .422$, $k = 23$, $R^2 = 0.031$; Grade: $\beta = 0.377$, $p = .058$, $k = 25$, $R^2 = 0.142$). This means that the correlations between RAN and reading accuracy were stable across the age (here, 3:7–12:2 years) or grade (here, kindergarten–Grade 6) range.

Correlations between RAN and reading fluency

Fourteen independent correlations comprising 1,356 children (M sample size = 96.86, $SD = 57.76$, range = 47–262) examined the relationship between RAN and reading fluency. The age of the samples ranged from 5:1 to 12:4 years. As shown in Figure 5, the overall mean correlation was large, $r = -.518$, 95% CI $[-.588, -.441]$, and significant, $z(13) = -11.214$, $p < .001$. The variation in correlations between studies was also significant and large, $Q(13) = 41.675$, $p < .001$, $I^2 = 68.806$. After removing outliers in the sensitivity analysis, the overall correlation ranged from $r = -.539$, 95% CI $[-.501, -.470]$, to $r = -.500$, 95% CI $[-.568, -.424]$. The funnel plot indicated no publication bias in the results. As shown in Table 4, dialect did not reliably explain variation in the correlation magnitude. In contrast, the test type was a significant moderator variable. Studies using graphological RAN tests produced significantly larger correlations than studies using nongraphological RAN tests. Age ($\beta = -0.011$, $p = .986$, $k = 5$, $R^2 < 0.001$) and grade ($\beta = 0.474$, $p = .236$, $k = 7$, $R^2 = 0.224$) had no significant impact on the size of the correlations. This means that the correlations between

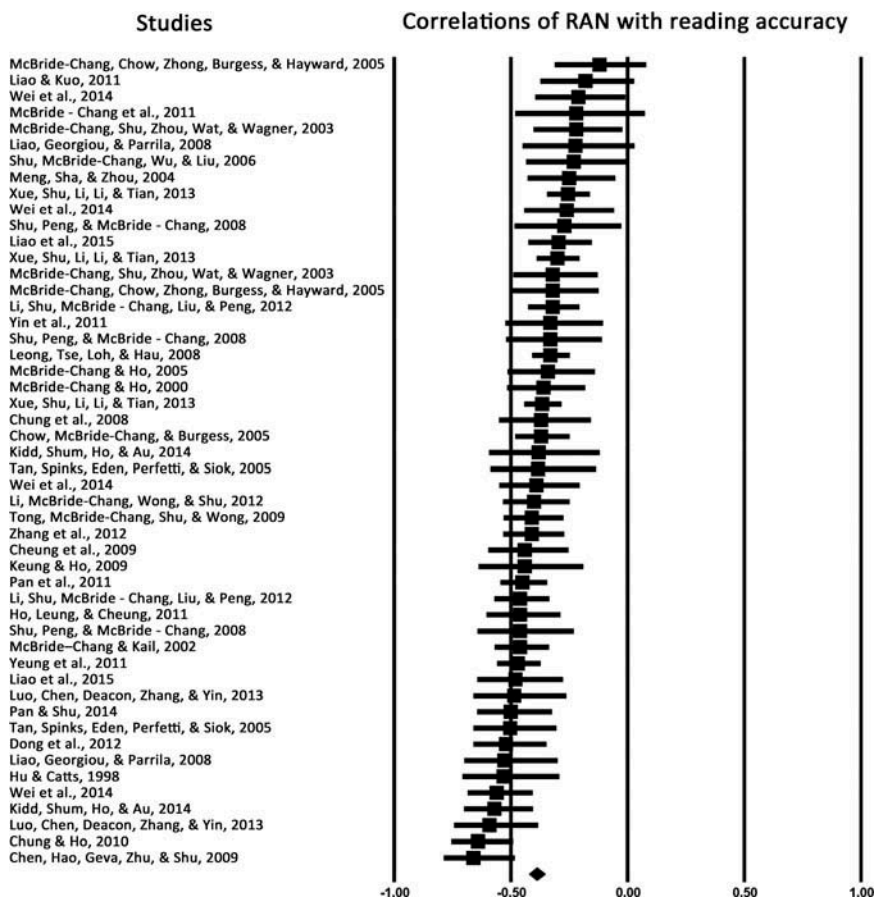


Figure 4. Overall average correlation (displayed by ♦) and correlation with confidence interval for each study correlating RAN and reading accuracy.

Table 3. Number of correlations, correlation size, 95% CI, heterogeneity statistics, difference in r between categories and significance test of categories for moderators of the relationship between RAN and reading accuracy.

Moderator variable	No. of correlations (k)	Correlation (r)	95% CI	Heterogeneity (I^2)	Difference in r (highest–lowest category)	Significance test of difference (Q test)
Test type						
Graphological	41	–.391**	–.43, –.36	73.73**		
Nongraphological	19	–.300**	–.35, –.56	5.03	.091	.002**
Dialect						
Cantonese	19	–.399**	–.40, –.35	49.01**		
Mandarin	31	–.379**	–.42, –.34	54.05**	.020	.552

Note. CI = confidence interval; I^2 = the proportion of total variation between the effect sizes that are caused by real heterogeneity rather than chance; r = correlation size (Pearson's r) for subsets of studies belonging to different categories of the moderator variable

** $p < .01$.

RAN and reading fluency were stable across the age (here, 7:5–8:9 years) or grade (here, Grades 1–6). Finally, the correlations between RAN and reading fluency were significantly stronger than the corresponding ones with reading accuracy, $Q(1) = 9.227$, $p = .002$.

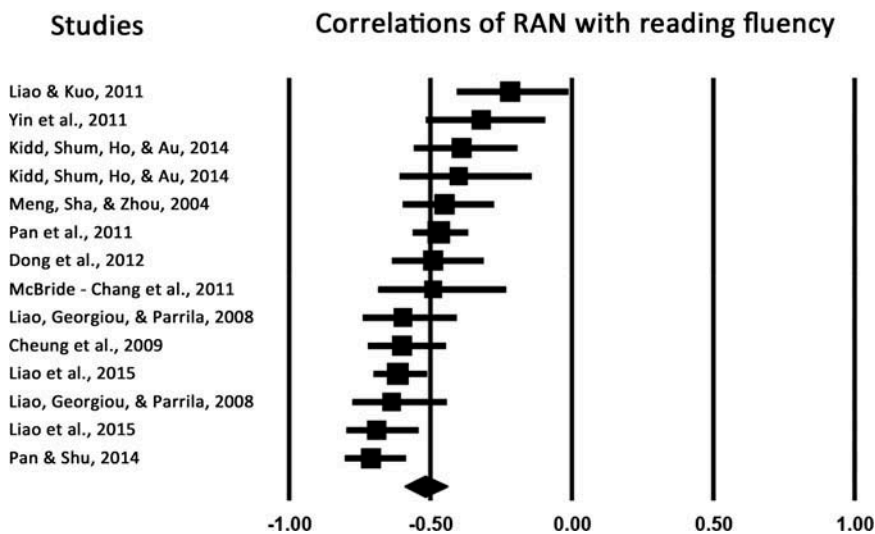


Figure 5. Overall average correlation (displayed by ♦) and correlation with confidence interval for each study correlating RAN and reading fluency.

Table 4. Number of correlations, correlation size, 95% CI, heterogeneity statistics, difference in r between categories and significance test of categories for moderators of the relationship between RAN and reading fluency.

Moderator variable	No. of correlations (k)	Correlation (r)	95% CI	Heterogeneity (I^2)	Difference in r (highest–lowest category)	Significance test of difference (Q test)
Test type						
Graphological	13	–.525**	–.60, –.45	70.39**		
Nongraphological	6	–.374**	–.46, –.28	15.30	.151	.012*
Dialect						
Cantonese	3	–.476**	–.61, –.32	50.62**		
Mandarin	11	–.529**	–.61, –.44	73.00**	.053	.533

Note. CI = confidence interval; I^2 = the proportion of total variation between the effect sizes that are caused by real heterogeneity rather than chance; r = correlation size (Pearson's r) for subsets of studies belonging to different categories of the moderator variable.

* $p < .05$. ** $p < .01$.

Discussion

The purpose of this meta-analysis was to examine the magnitude of the association between phonological awareness, RAN, and Chinese reading accuracy and fluency. This is important for generalizability purposes because most previous meta-analyses included studies conducted predominantly in alphabetic languages. Chinese, as a logographic script, differs notably from alphabetic languages such as English in orthography, phonology, and semantics (Hanley, 2005; see also Tan, Laird, Li, & Fox, 2005; for differences between alphabetic languages and Chinese at the neural level). Thus, research on the Chinese language is important to advance our understanding of the universality or particularity of the cognitive processes that underlie reading development.

The findings of our meta-analysis showed first that phonological awareness correlated significantly with reading accuracy ($r = .36$) and reading fluency ($r = .39$). However, the mean correlation coefficient between phonological awareness and reading accuracy in Chinese is significantly lower (based on the 95% CI) than the ones reported in previous meta-analyses with studies conducted in alphabetic languages (.57 in Melby-Lervåg et al.'s meta-analysis, .46 in Scarborough's meta-analysis, and .41 in Swanson et al.'s meta-analysis). This reinforces the argument put forward by some

researchers that the effects of phonological awareness in Chinese reading may not be as strong as in alphabetic languages, in which there is a more transparent link between graphemes and phonemes (e.g., Huang & Hanley, 1995; Shu et al., 2006; Taylor, 2002; Xue et al., 2013).

Although the correlations between phonological awareness and reading accuracy in Chinese are lower than the corresponding ones in alphabetic languages, we should not dismiss the fact that they are still significant. An explanation for this could be that phonology is activated in Chinese the same way it does in alphabetic languages (e.g., Perfetti & Liu, 2005; Perfetti, Zhang, & Berent, 1992). This alone may result in significant correlations between phonological awareness tasks and character recognition. However, even though phonology is activated in Chinese reading, it does not contribute to character recognition the same way it does in alphabetic languages in which children rely on grapheme–phoneme correspondences to read unfamiliar words (at least at the initial phases of reading development). This difference may account for the observed differences in the size of the correlations between phonological awareness and reading in Chinese and in alphabetic languages. An alternative explanation for the significant correlation between phonological awareness and Chinese reading may be that the small number of the phonetic compound characters that can be read accurately using the phonetic radical (Chung & Leung, 2008) is good enough to produce significant correlations with phonological awareness.

Our findings also showed that the relationship between phonological awareness and Chinese reading did not vary as a function of test type or dialect. This suggests first that even complex phonological awareness tasks such as tone awareness or phoneme deletion can produce correlations with reading that are not significantly different from more conventional phonological awareness tasks in Chinese (e.g., syllable deletion, onset detection; Newman, Tardif, Huang, & Shu, 2011). Second, it suggests that exposure of children to phonetic coding systems (e.g., Pinyin) may give an advantage to Mandarin- over Cantonese-speaking children in phonological awareness (e.g., Huang & Hanley, 1995; Leong et al., 2005; McBride-Chang et al., 2004), but it does not translate to stronger associations between phonological awareness and reading in Mandarin. This is certainly an intriguing finding that warrants further investigation in future studies. However, a possible explanation may be that even though children in Hong Kong are not directly exposed to a phonetic coding system, they still get some training in phonological awareness through second-language learning (English) that starts from an early age. In line with this, Chen et al. (2004) found that bilingual Chinese children had better phonological awareness skills than monolingual Chinese children.

Our results also showed that RAN was significantly related with reading accuracy ($r = -.38$) and reading fluency ($r = -.51$). Notably, the correlations are similar to those reported in previous meta-analyses for alphabetic languages (Araújo et al., 2015; Scarborough, 1998; Swanson et al., 2003). This has implications in regards to the theoretical accounts that have been used to explain the RAN–reading relationship (Pan et al., 2011; Shu et al., 2006; Xue et al., 2013; Yeung et al., 2011). We hypothesized that if the “arbitrary connections” or the “visual-complexity” hypotheses were correct, we should observe higher correlations between RAN and reading in Chinese than in alphabetic languages. Certainly, our findings do not support either hypothesis. However, we acknowledge that the correlation between RAN and reading fluency is based on a smaller number of studies ($n = 14$) than in Araújo et al.’s (2015) study ($n = 55$) and, as such, our finding should be viewed with some caution. Nonetheless, our finding suggests that reading fluency is an important component that should be included in the assessment of reading in Chinese.

Test type moderated the relationship between RAN and reading such that higher correlations were observed when graphological RAN tasks were used. This finding is in line with those of previous studies in alphabetic languages (e.g., Bowey, McGuigan, & Ruschena, 2005; Compton, 2003; Georgiou et al., 2008; Wolf, Bally, & Morris, 1986) and suggests that graphological RAN tasks reflect more closely the processes taking place in reading. Neuroimaging studies have also shown a greater overlap between the neural networks of reading and graphological RAN than nongraphological RAN (e.g., Cummine, Szepesvari, Chouinard, & Georgiou, 2015; Misra, Katzir, Wolf, & Roldrack, 2004). In addition, the correlations between RAN and reading fluency were significantly

higher than the corresponding ones with reading accuracy. This is in line with the finding of previous studies in Chinese (e.g., Liao et al., 2015; Liao et al., 2008; Pan et al., 2011), as well as in alphabetic languages (e.g., Georgiou, Parrila, & Kirby, 2009; Moll et al., 2013; Savage & Frederickson, 2005; Vaessen et al., 2010) and verifies one of the premises of the double-deficit hypothesis, namely, that phonological awareness and RAN should predict different kinds of reading outcomes (RAN making a greater contribution to reading fluency than reading accuracy).

There might be three explanations for the significantly larger association between RAN and reading fluency than reading accuracy. First, both RAN and reading fluency measures require speeded responses, and this may boost their relationship. Although this is a plausible explanation, we cannot dismiss the evidence showing that RAN continues to predict reading even after controlling for speed of processing measures (e.g., Bowey et al., 2005; Georgiou et al., 2009). Second, some researchers have argued that the speed with which high-frequency words are retrieved from long-term memory is similar to the speed individual letters or digits are retrieved from long-term memory (Georgiou et al., 2009; Protopapas et al., 2013). In other words, the retrieval of these high-frequency words has become as automatic as the retrieval of single digits or letters. The ability to retrieve whole-word representations from long-term memory is obviously more important for reading fluency than reading accuracy. Finally, eye-movement studies have shown that serial RAN tasks allows individuals to take advantage of paravofeal information when naming the stimuli (e.g., Jones, Ashby, & Branigan, 2013; Yan, Pan, Laubrock, Kliegl, & Shu, 2013). Thus, although participants articulate a stimulus in position n , they have already accessed the phonological representation of stimulus in $n + 1$, and visually scanned stimulus in $n + 2$. This is also possible in reading fluency tasks (all the words are usually printed on one page) but not in reading accuracy (the words are usually presented one at a time).

Age or grade did not moderate the relationship between RAN and reading (although the RAN–reading accuracy correlations appeared to increase across grades). The effects of time when RAN and reading are assessed has created a lot of controversy in the literature. In a longitudinal study from kindergarten to Grade 5, Kirby et al. (2003) found that the relative contribution of RAN to reading accuracy increased with age. Other studies have supported this trend (e.g., Batnini & Uno, 2015; Landerl & Wimmer, 2008; Vaessen et al., 2010). In contrast, Torgesen, Wagner, Rashotte, Burgess, and Hecht (1997) found that RAN had its primary influence on reading in Grades 1 and 2 and that its influence diminishes in Grades 3 to 5 (see also Georgiou et al., 2009; Roman, Kirby, Parrila, Wade-Woolley, & Deacon, 2009, for similar findings). What seems to have been missed in this debate is the role the type of reading outcome may play. A closer examination at Torgesen et al.'s (1997) results, for example, shows that the correlations between RAN and reading accuracy measures dropped across time but the correlations between RAN and reading fluency measures did not. Araújo and colleagues (2015) also found that the RAN–reading accuracy relationship declined across grades but the RAN–reading fluency remained rather stable. The absence of a significant effect of age or grade in our meta-analysis may reflect the fact that Chinese children are introduced in every grade of the elementary school to a large number of new characters that cannot be read accurately unless the teacher provides the correct pronunciation (see Shu et al., 2003). This is not the case in alphabetic languages in which children learn the mechanism of decoding in Grade 1 and can successfully read new words (particularly in consistent orthographies) with the selection and application of different reading strategies.

Some limitations of the present study are worth mentioning. First, some studies have used RAN characters as one of their measures. However, one may argue that if children know the characters in RAN, they likely know how to read. This leads to a vicious circle in which reading is used to predict reading. To the extent this is true, then the inclusion of these studies in the meta-analysis may have inflated the correlations between RAN and reading in Chinese and may partly account for the significantly higher correlations between graphological RAN and reading than between nongraphological RAN and reading. Second, our meta-analysis focused on the relationships of RAN with word reading and did not assess the possible connections of phonological awareness and RAN with

reading comprehension. Finally, we deliberately chose to include studies in our meta-analysis that assessed both phonological awareness and RAN in the same study. We made this decision in order to gain a better control over the possible effects of confounding variables (e.g., sample characteristics) in the size of the correlations. However, this reduced the number of studies that were considered in the meta-analysis.

To conclude, the findings of our meta-analysis in Chinese extend those of previous meta-analyses that included studies conducted predominately in alphabetic languages. We have shown that both phonological awareness and RAN are significant correlates of Chinese reading accuracy and fluency. Viewed together with the findings of previous meta-analyses, our findings suggest that, despite significant differences between writing systems (logographic, phonetic), phonological awareness and RAN are likely universal correlates of reading. However, whether they are unique predictors of Chinese reading depends on what other processing skills (e.g., morphological awareness, copying skills) are taken into account when predicting reading.

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Appendix

Table A1. Information on the studies included in the meta-analysis.

Study name	Task	Accuracy/Fluency	Subgroup	r	Sample	Dialect	Age	Type
Chen, Hao, Geva, Zhu, & Shu, 2009	PA	Accuracy		.34	59	Mandarin	7.34	Simple
	RAN	Accuracy		-.66	59	Mandarin	7.34	Graphological
Cheung et al., 2009	Morphological	Accuracy		.54	59	Mandarin	7.34	Compound analogy
	PA	Accuracy		.52	90	Cantonese	9.92	Combined
	RAN	Accuracy		-.44	90	Cantonese	9.92	Graphological
	PA	Fluency		.33	90	Cantonese	9.92	Combined
	RAN	Fluency		-.60	90	Cantonese	9.92	Graphological
Chow, McBride-Chang, & Burgess, 2005	PA	Accuracy		.42	227	Cantonese	4.88	Simple
	RAN	Accuracy		-.37	227	Cantonese	4.88	Graphological
	PA	Accuracy		.39	78	Cantonese	8.7	Combined
	RAN	Accuracy		-.37	78	Cantonese	8.7	Graphological
	Morphological	Accuracy		.50	78	Cantonese	8.7	Construction and production
Chung & Ho, 2010	Orthographic	Accuracy		.23	78	Cantonese	8.7	Lexical decision
	PA	Accuracy		.40	84	Cantonese	9.05	
	RAN	Accuracy		-.64	84	Cantonese	9.05	Graphological
	Morphological	Accuracy		.67	84	Cantonese	9.05	
	Orthographic	Accuracy		.65	84	Cantonese	9.05	
Dong et al., 2012	PA	Accuracy		.62	90	Mandarin	10.31	Complex
	RAN	Accuracy		-.52	90	Mandarin	10.31	Graphological
	Morphological	Accuracy		.46	90	Mandarin	10.31	Production
	Orthographic	Accuracy		.28	90	Mandarin	10.31	Lexical decision
	PA	Fluency		.40	90	Mandarin	10.31	Complex
	RAN	Fluency		-.49	90	Mandarin	10.31	Graphological
	Morphological	Fluency		.40	90	Mandarin	10.31	Production
	Orthographic	Fluency		.33	90	Mandarin	10.31	Lexical decision
	PA	Accuracy		.51	101	Cantonese	6.04	Simple
	RAN	Accuracy		-.46	101	Cantonese	6.04	Combined
Hu & Catts, 1998	Morphological	Accuracy		.35	101	Cantonese	6.04	Combined
	PA	Accuracy		.68	50	Mandarin	6.83	Simple
	RAN	Accuracy		-.53	50	Mandarin	6.83	Nongraphological
Keung & Ho, 2009	PA	Accuracy		.17	53	Cantonese	8.17	Simple
	RAN	Accuracy		-.44	53	Cantonese	8.17	Nongraphological
	Orthographic	Accuracy		-.26	53	Cantonese	8.17	Lexical decision
	PA	Accuracy	Kindergarten	.45	53	Cantonese	5.23	Simple
	RAN	Accuracy	Kindergarten	-.38	53	Cantonese	5.23	Nongraphological
Kidd, Shum, Ho, & Au, 2015	PA	Fluency	Kindergarten	.40	53	Cantonese	5.23	Simple
	RAN	Fluency	Kindergarten	-.40	53	Cantonese	5.23	Nongraphological
	PA	Accuracy	Primary	.34	87	Cantonese	8.3	Simple
	RAN	Accuracy	Primary	-.57	87	Cantonese	8.3	Graphological
	PA	Fluency	Primary	.14	87	Cantonese	8.3	Simple
	RAN	Fluency	Primary	-.39	87	Cantonese	8.3	Graphological

(Continued)

Table A1. (Continued).

Study name	Task	Accuracy/Fluency	Subgroup	r	Sample	Dialect	Age	Type
Leong, Tse, Loh, & Hau, 2008	PA	Accuracy		.33	518	Cantonese	10.22	Complex
	RAN	Accuracy		.33	518	Cantonese	10.22	Graphological
	PA	Accuracy		.20	141	Cantonese	8.1	Combined
Li, McBride-Chang, Wong, & Shu, 2012	RAN	Accuracy		-.40	141	Cantonese	8.1	Graphological
	PA	Accuracy	Kindergarten	.07	184	Mandarin	5.34	Simple
	RAN	Accuracy	Kindergarten	-.46	184	Mandarin	5.34	Graphological
Li, Shu, McBride-Chang, Liu, & Peng, 2012	Morphological	Accuracy	Kindergarten	.35	184	Mandarin	5.34	Construction
	Orthographic	Accuracy	Kindergarten	.05	184	Mandarin	5.34	Position and ill-formed component
	PA	Accuracy	Primary	.32	273	Mandarin	7.89	Simple
	PA	Accuracy	Primary	.19	273	Mandarin	7.89	Complex
	RAN	Accuracy	Primary	-.32	273	Mandarin	7.89	Graphological
	Morphological	Accuracy	Primary	.34	273	Mandarin	7.89	Production
	Orthographic	Accuracy	Primary	.25	273	Mandarin	7.89	Position and ill-formed component
	PA	Accuracy		.29	93	Mandarin	12.3	Simple
	RAN	Accuracy		-.18	93	Mandarin	12.3	Graphological
	RAN	Accuracy		-.18	93	Mandarin	12.3	Nongraphological
Liao & Kuo, 2011	PA	Fluency		.20	93	Mandarin	12.3	Simple
	RAN	Fluency		-.22	93	Mandarin	12.3	Graphological
	RAN	Fluency		-.23	93	Mandarin	12.3	Nongraphological
	RAN	Accuracy	Grade 2	-.30	182	Mandarin	8.15	Graphological
	RAN	Fluency	Grade 2	-.62	182	Mandarin	8.15	Graphological
	RAN	Accuracy	Grade 2	-.48	72	Mandarin	8.19	Graphological
	RAN	Accuracy	Grade 2	-.27	72	Mandarin	8.19	Nongraphological
	PA	Accuracy	Grade 2	.43	72	Mandarin	8.19	Simple
	RAN	Fluency	Grade 2	-.69	72	Mandarin	8.19	Graphological
	RAN	Fluency	Grade 2	-.38	72	Mandarin	8.19	Nongraphological
Liao, Georgiou, & Parrila, 2008	PA	Fluency	Grade 2	.69	72	Mandarin	8.19	Simple
	PA	Accuracy	Grade 2	.32	63	Mandarin	8	Simple
	PA	Accuracy	Grade 2	.45	63	Mandarin	8	Complex
	RAN	Accuracy	Grade 2	-.21	63	Mandarin	8	Nongraphological
	RAN	Accuracy	Grade 2	-.22	63	Mandarin	8	Graphological
	Orthographic	Accuracy	Grade 2	.28	63	Mandarin	8	Noncharacter
	PA	Fluency	Grade 2	.44	63	Mandarin	8	Simple
	PA	Fluency	Grade 2	.47	63	Mandarin	8	Complex
	RAN	Fluency	Grade 2	-.53	63	Mandarin	8	Nongraphological
	RAN	Fluency	Grade 2	-.60	63	Mandarin	8	Graphological
	Orthographic	Fluency	Grade 2	.23	63	Mandarin	8	Noncharacter
	PA	Accuracy	Grade 4	.67	54	Mandarin	10.01	Simple
	PA	Accuracy	Grade 4	.51	54	Mandarin	10.01	Complex
	RAN	Accuracy	Grade 4	-.25	54	Mandarin	10.01	Nongraphological
	RAN	Accuracy	Grade 4	-.53	54	Mandarin	10.01	Graphological

(Continued)

Table A1. (Continued).

Study name	Task	Accuracy/Fluency	Subgroup	r	Sample	Dialect	Age	Type
Luo, Chen, Deacon, Zhang, & Yin, 2013	Orthographic	Accuracy	Grade 4	.54	54	Mandarin	10.01	Noncharacter
	PA	Fluency	Grade 4	.34	54	Mandarin	10.01	Simple
	PA	Fluency	Grade 4	.34	54	Mandarin	10.01	Complex
	RAN	Fluency	Grade 4	-.47	54	Mandarin	10.01	Nongraphological
	RAN	Fluency	Grade 4	-.64	54	Mandarin	10.01	Graphological
	Orthographic	Fluency	Grade 4	.18	54	Mandarin	10.01	Noncharacter
	PA	Accuracy	Kindergarten1	.47	55	Mandarin	5.16	Combined
	PA	Accuracy	Kindergarten1	.69	26	Mandarin	5.16	Simple
	PA	Accuracy	Kindergarten1	-.09	26	Mandarin	5.16	Complex
	PA	Accuracy	Kindergarten2	-.17	26	Mandarin	6.16	Simple
	PA	Accuracy	Kindergarten2	.45	26	Mandarin	6.16	Complex
	RAN	Accuracy	Kindergarten2	-.59	55	Mandarin	5.66	Graphological
	PA	Accuracy		.34	61	Mandarin	7.68	Combined
	PA	Accuracy	Primary1	.32	30	Mandarin	7.18	Simple
	PA	Accuracy	Primary1	.38	30	Mandarin	7.18	Complex
McBride-Chang, Chow, Zhong, Burgess, & Hayward, 2005	PA	Accuracy	Primary2	-.05	31	Mandarin	8.18	Simple
	PA	Accuracy	Primary2	-.10	31	Mandarin	8.18	Complex
	RAN	Accuracy		-.49	61	Mandarin	7.68	Graphological
	PA	Accuracy		.37	101	Cantonese	5.32	Simple
	RAN	Accuracy		-.12	101	Cantonese	5.32	Nongraphological
	PA	Accuracy		.45	95	Mandarin	4.93	Simple
	RAN	Accuracy		-.32	95	Mandarin	4.93	Nongraphological
	PA	Accuracy		.51	109	Cantonese	4.02	Simple
	RAN	Accuracy		-.23	109	Cantonese	4.02	Nongraphological
	RAN	Accuracy		-.36	109	Cantonese	4.02	Graphological
	PA	Accuracy		.11	90	Cantonese	4.02	Simple
	RAN	Accuracy		-.34	90	Cantonese	4.02	Nongraphological
	RAN	Accuracy		-.34	90	Cantonese	5.85	Graphological
	PA	Accuracy		.63	190	Cantonese	5.07	Simple
	RAN	Accuracy		-.46	190	Cantonese	5.07	Graphological
McBride-Chang et al., 2011	RAN	Accuracy		-.35	190	Cantonese	5.07	Nongraphological
	RAN	Accuracy		-.22	47	Mandarin	5.08	Graphological
	PA	Accuracy		.26	47	Mandarin	5.08	Simple
	PA	Accuracy		.56	47	Mandarin	5.08	Complex
	Morphological	Accuracy		.25	47	Mandarin	5.08	Construction
	RAN	Fluency		-.49	47	Mandarin	5.08	Graphological
	PA	Fluency		.09	47	Mandarin	5.08	Simple
	PA	Fluency		.37	47	Mandarin	5.08	Complex
	Morphological	Fluency		.12	47	Mandarin	5.08	Construction
	PA	Accuracy	Kindergarten	.36	100	Cantonese	5.25	Simple
	PA	Accuracy	Kindergarten	.16	100	Cantonese	5.25	Complex
McBride-Chang, Shu, Zhou, Wat, & Wagner, 2003								

(Continued)

Table A1. (Continued).

Study name	Task	Accuracy/Fluency	Subgroup	r	Sample	Dialect	Age	Type
Meng, Sha, & Zhou, 2004	RAN	Accuracy	Kindergarten	-.32	100	Cantonese	5.25	Graphological
	RAN	Accuracy	Kindergarten	-.35	100	Cantonese	5.25	Nongraphological
	Morphological	Accuracy	Kindergarten	.51	100	Cantonese	5.25	Construction
	PA	Accuracy	Primary	.14	100	Cantonese	7.25	Simple
	PA	Accuracy	Primary	.33	100	Cantonese	7.25	Complex
	RAN	Accuracy	Primary	-.22	100	Cantonese	7.25	Graphological
	RAN	Accuracy	Primary	-.33	100	Cantonese	7.25	Nongraphological
	Morphological	Accuracy	primary	.40	100	Cantonese	7.25	Construction
	PA	Accuracy		.53	100	Mandarin	8.75	Simple
	RAN	Accuracy		-.25	100	Mandarin	8.75	Graphological
Pan et al., 2011	RAN	Accuracy		-.18	100	Mandarin	8.75	Nongraphological
	PA	Fluency		.44	100	Mandarin	8.75	Simple
	RAN	Fluency		-.45	100	Mandarin	8.75	Graphological
	RAN	Fluency		-.32	100	Mandarin	8.75	Nongraphological
	PA	Accuracy		.41	262	Mandarin	5.4	Simple
	RAN	Accuracy		-.45	262	Mandarin	5.4	Graphological
	PA	Fluency		.39	262	Mandarin	7.4	Simple
	RAN	Fluency		-.47	262	Mandarin	7.4	Graphological
	PA	Accuracy		.64	90	Mandarin	9.31	Complex
	RAN	Accuracy		-.50	90	Mandarin	9.31	Graphological
Pan & Shu, 2014	PA	Fluency		.54	90	Mandarin	9.31	Complex
	RAN	Fluency		-.71	90	Mandarin	9.31	Graphological
	Morphological	Accuracy		.64	77	Mandarin	11.5	Production
	RAN	Accuracy		-.23	77	Mandarin	11.5	Graphological
	RAN	Accuracy		-.30	77	Mandarin	11.5	Nongraphological
	PA	Accuracy		.46	77	Mandarin	11.5	Complex
	PA	Accuracy		.47	77	Mandarin	11.5	Simple
	PA	Accuracy	Kindergarten1	.31	60	Mandarin	40-47	Simple
	PA	Accuracy	Kindergarten1	.28	60	Mandarin	40-47	Complex
	RAN	Accuracy	Kindergarten1	-.46	60	Mandarin	40-47	Nongraphological
Shu, Peng, & McBride-Chang, 2008	PA	Accuracy	Kindergarten2	.22	76	Mandarin	48-59	Simple
	PA	Accuracy	Kindergarten2	.31	76	Mandarin	48-59	Complex
	RAN	Accuracy	Kindergarten2	-.33	76	Mandarin	48-59	Nongraphological
	PA	Accuracy	Kindergarten3	.30	66	Mandarin	60-78	Simple
	PA	Accuracy	Kindergarten3	.25	66	Mandarin	60-78	Complex
	RAN	Accuracy	Kindergarten3	-.27	66	Mandarin	60-78	Nongraphological
	PA	Accuracy	Beginning	.31	58	Mandarin	8.18	Complex
	RAN	Accuracy	Beginning	-.38	58	Mandarin	8.18	Graphological
	PA	Accuracy	Intermediate	.33	73	Mandarin	9.74	Complex
	RAN	Accuracy	Intermediate	-.50	73	Mandarin	9.74	Graphological
Tong, McBride-Chang, Shu, & Wong, 2009	RAN	Accuracy		-.41	171	Cantonese	6.13	Graphological

(Continued)

Table A1. (Continued).

Study name	Task	Accuracy/Fluency	Subgroup	r	Sample	Dialect	Age	Type
Wei et al., 2014	PA	Accuracy		.26	171	Cantonese	6.13	Combined
	Morphological	Accuracy		.52	171	Cantonese	6.13	Construction
	Orthographic	Accuracy		.42	171	Cantonese	6.13	Lexical decision
	PA	Accuracy	Preschool	.29	101	Mandarin	6.06	Simple
	PA	Accuracy	Preschool	.45	101	Mandarin	6.06	Complex
	RAN	Accuracy	Preschool	-.56	101	Mandarin	6.06	Graphological
	Morphological	Accuracy	Preschool	.29	101	Mandarin	6.06	Average
	Orthographic	Accuracy	Preschool	.59	101	Mandarin	6.06	Average
	PA	Accuracy	Grade 1	.12	94	Mandarin	7.09	Simple
	PA	Accuracy	Grade 1	.19	94	Mandarin	7.09	Complex
	RAN	Accuracy	Grade 1	-.26	94	Mandarin	7.09	Graphological
	Morphological	Accuracy	Grade 1	.29	94	Mandarin	7.09	Average
	Orthographic	Accuracy	Grade 1	.40	94	Mandarin	7.09	Average
	PA	Accuracy	Grade 2	.42	98	Mandarin	8	Simple
	PA	Accuracy	Grade 2	.31	98	Mandarin	8	Complex
	RAN	Accuracy	Grade 2	-.39	98	Mandarin	8	Graphological
Xue, Shu, Li, Li, & Tian, 2013	Morphological	Accuracy	Grade 2	.33	98	Mandarin	8	Average
	Orthographic	Accuracy	Grade 2	.21	98	Mandarin	8	Average
	PA	Accuracy	Grade 3	.15	98	Mandarin	9.11	Simple
	PA	Accuracy	Grade 3	.30	98	Mandarin	9.11	Complex
	RAN	Accuracy	Grade 3	-.21	98	Mandarin	9.11	Graphological
	Morphological	Accuracy	Grade 3	.28	98	Mandarin	9.11	Average
	Orthographic	Accuracy	Grade 3	.10	98	Mandarin	9.11	Average
	RAN	Accuracy	Grade 2	-.30	408	Mandarin	8.3	Graphological
	PA	Accuracy	Grade 2	.29	408	Mandarin	8.3	Complex
	Morphological	Accuracy	Grade 2	.24	408	Mandarin	8.3	Production
	Orthographic	Accuracy	Grade 2	.25	408	Mandarin	8.3	Lexical decision
	RAN	Accuracy	Grade 4	-.26	428	Mandarin	10.31	Graphological
	PA	Accuracy	Grade 4	.31	428	Mandarin	10.31	Complex
	Morphological	Accuracy	Grade 4	.32	428	Mandarin	10.31	Production
	Orthographic	Accuracy	Grade 4	.23	428	Mandarin	10.31	Lexical decision
	RAN	Accuracy	Grade 6	-.37	496	Mandarin	12.14	Graphological
Yeung et al., 2011	PA	Accuracy	Grade 6	.38	496	Mandarin	12.14	Complex
	Morphological	Accuracy	Grade 6	.27	496	Mandarin	12.14	Production
	Orthographic	Accuracy	Grade 6	.28	496	Mandarin	12.14	Lexical decision
	PA	Accuracy		.21	290	Cantonese	6.6	Simple
	RAN	Accuracy		-.47	290	Cantonese	6.6	Graphological
	Morphological	Accuracy		.48	290	Cantonese	6.6	Identification
Yin et al., 2011	Orthographic	Accuracy		.48	290	Cantonese	6.6	Orthography
	PA	Accuracy		.03	73	Mandarin	8.5	Simple
	PA	Accuracy		.41	73	Mandarin	8.5	Complex

(Continued)

Table A1. (Continued).

Study name	Task	Accuracy/Fluency	Subgroup	r	Sample	Dialect	Age	Type
Zhang et al., 2012	RAN	Accuracy		-.33	73	Mandarin	8.5	Graphological
	PA	Fluency		.05	73	Mandarin	8.5	Simple
	PA	Fluency		.26	73	Mandarin	8.5	Complex
	RAN	Fluency		-.32	73	Mandarin	8.5	Graphological
	Morphological	Accuracy		.42	164	Cantonese	7.37	Construction
	Orthographic	Accuracy		.28	164	Cantonese	7.37	Orthography-semantic awareness
	RAN	Accuracy		-.41	164	Cantonese	7.37	Graphological
	PA	Accuracy		.34	164	Cantonese	7.37	Combined

Note. PA = phonological awareness; RAN = rapid automatized naming.